

SentinelAI-X: Interface Discovery Simple Explanation

Module Purpose

interface_discovery.py ka kaam system ke available network interfaces ko detect karna hai. Ye module packet_capture.py se pehle use hoga, taaki user ko pata chale kaunsi interface par traffic capture karna hai.

Main Imports

argparse CLI arguments ke liye, logging structured logs ke liye, dataclass clean data model ke liye, aur scapy.all.IFACES network interfaces read karne ke liye use hota hai.

NetworkInterface Class

Ye dataclass ek interface ka normalized data store karti hai: name, description, index, mac_address, aur ip_address. frozen=True data ko immutable banata hai, slots=True memory efficient banata hai.

JsonLogFormatter Class

Ye logs ko JSON format me convert karta hai. Cybersecurity tools me structured logs useful hote hain kyunki SIEM ya log analyzers easily parse kar sakte hain.

configure_logging() Function

Ye root logger setup karta hai aur JsonLogFormatter attach karta hai. User --log-level se DEBUG, INFO, ERROR jaise levels choose kar sakta hai.

InterfaceDiscovery Class

Ye reusable main class hai. Future packet capture modules direct Scapy use karne ke bajay is class se clean NetworkInterface objects le sakte hain.

discover() Function

Ye IFACES.values() se raw Scapy interfaces leta hai, har interface ko _normalize_interface() se clean format me convert karta hai, aur list return karta hai. Error aane par InterfaceDiscoveryError raise hoti hai.

get_interface_by_name() Function

Ye function interface name se specific interface search karta hai. casefold() use hota hai taaki comparison case-insensitive ho.

render_table() Function

Ye discovered interfaces ko readable table me print karta hai. Missing MAC/IP/index ke liye N/A show hota hai.

Helper Functions

_safe_string(), _safe_optional_string(), aur _safe_int() raw Scapy attributes ko safely read karte hain. _extract_ip_address() platform differences handle karta hai, kyunki Scapy IP info kabhi ip aur kabhi ips attribute me de sakta hai.

main() Flow

main() CLI args parse karta hai, logging configure karta hai, InterfaceDiscovery object banata hai, interfaces discover karta hai, aur final table print karta hai. Success par 0 aur failure par 1 return hota hai.

Architecture Use

Sensor Module flow: interface_discovery.py pehle interface choose karne me help karega. Uske baad packet_capture.py selected interface par packets capture karega, packet_parser.py parse karega, traffic_logger.py store karega, aur feature_extractor.py ML/security features nikalega.

Run Command

```
python sensor/interface_discovery.py --log-level INFO
```